CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

Chapter 4: Exploring Some Geometric Themes

Topic-wise short notes with solved examples

Two themes in geometry: fractals, which repeat the same pattern at smaller and smaller scales, and ways of visualising solids through nets, shortest paths, projections (front, top and side views) and isometric drawings.

1 Fractals

- A **fractal** is a self-similar shape: it shows the same pattern again and again at smaller and smaller scales.
- In nature: ferns, trees, clouds, coastlines, mountains and lightning.
- In art and architecture: the towers of temples such as the Kandariya Mahadev Temple at Khajuraho (about 1025 CE), patterns on Fulani wedding blankets of Nigeria, and prints by the artist M. C. Escher.
- A mathematical fractal is the result of repeating one geometric step for ever. Each stage of the repetition is called a **step**: Step 0, Step 1, Step 2, and so on.

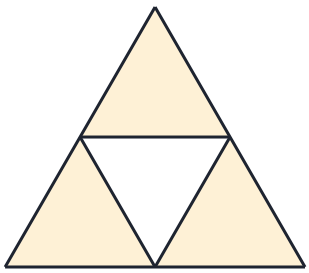
2 Sierpinski Carpet

- **Rule**: divide a square into 9 equal smaller squares and remove the central one. Repeat on each of the 8 squares that remain.
- Let R_n be the number of squares remaining and H_n the number of holes at Step n.
- Each remaining square produces 8 new squares: $R_{n+1} = 8R_n$, so $R_n = 8^n$.
- Each remaining square also produces one new hole: $H_{n+1} = H_n + R_n$, so $H_n = 1 + 8 + 8^2 + \ldots + 8^{n-1}$.
- Each step keeps $\frac{8}{9}$ of the area. Starting from area 1, the area at Step n is $\left(\frac{8}{9}\right)^n$.

Step	0	1	2	3
Squares remaining	1	8	64	512
Holes	0	1	9	73
Area remaining	1	$\frac{8}{9}$	$\frac{64}{81}$	$\frac{512}{729}$

3 Sierpinski Triangle

- **Rule**: join the mid-points of the sides of an equilateral triangle to get 4 congruent equilateral triangles, and remove the central one. Repeat on each of the 3 triangles that remain. It is also called the Sierpinski Gasket.



Step 1: the middle triangle is removed, three remain

- Triangles remaining at Step n: 3^n. Holes: $1 + 3 + 3^2 + \ldots + 3^{n-1}$. Area remaining: $\left(\frac{3}{4}\right)^n$ of the original.
- Why the four triangles are congruent: each corner triangle has two sides equal to half a side with a 60° angle between them, so it is equilateral with side half the original. The middle triangle then also has three sides of that length.

Example 3.1 How many triangles remain, and how many holes are there, at Step 4 of the Sierpinski Triangle?

SOLUTION

Triangles: $3^4 = 81$. Holes: $1 + 3 + 9 + 27 = 40$.

81 triangles and 40 holes

Example 3.2 What fraction of the original area remains at Step 3 of the Sierpinski Triangle?

SOLUTION

$\left(\frac{3}{4}\right)^3$

$\frac{27}{64}$

4 Koch Snowflake

- **Rule**: start with an equilateral triangle. Divide each side into 3 equal parts, raise an equilateral triangle on the middle part and remove that middle part. Repeat on every side of the new shape. (Described by Helge von Koch in 1904.)
- Each side is replaced by 4 sides, each one-third as long.
- Number of sides at Step n: 3×4^n. Length of each side (starting from 1): $\frac{1}{3^n}$.
- Perimeter at Step n: $3 \times \left(\frac{4}{3}\right)^n$. The perimeter keeps growing without limit, although the shape stays inside a fixed region.

Example 4.1 Find the number of sides and the perimeter of the Koch Snowflake at Step 2, starting from a triangle of side 1 unit.

SOLUTION

Sides: $3 \times 4^2 = 48$, each of length $\frac{1}{9}$.

48 sides; perimeter $\frac{48}{9} = 5\frac{1}{3}$ units

5 Solids: faces, edges, vertices

- **Faces** are the flat surfaces of a solid. **Edges** are the line segments where faces meet. **Vertices** are the points where edges meet. A cuboid has 6 faces, 12 edges and 8 vertices.
- A **prism** has two congruent polygons as opposite faces, joined by parallelograms. It is named after the polygon: triangular prism, pentagonal prism, and so on.
- A **pyramid** has a polygon as base and a point (apex) joined to every vertex of the base. A triangular pyramid is a **tetrahedron**.
- An **octahedron** is two square pyramids joined at their bases: 8 triangular faces. A **dodecahedron** has 12 pentagonal faces.

Solid	Faces	Edges	Vertices
Prism on an n-sided polygon	$n+2$	$3n$	$2n$
Pyramid on an n-sided polygon	$n+1$	$2n$	$n+1$
Cube or cuboid	6	12	8
Tetrahedron	4	6	4
Octahedron	8	12	6

- For all of these, Faces + Vertices − Edges = 2.

Example 5.1 A prism has two 10-sided polygons as its ends. Find its faces, edges and vertices.

SOLUTION

Faces: $10 + 2$. Edges: 3×10. Vertices: 2×10.

12 faces, 30 edges, 20 vertices

Example 5.2 A pyramid has a 10-sided base. Find its faces, edges and vertices.

SOLUTION

11 faces, 20 edges, 11 vertices

6 Profiles

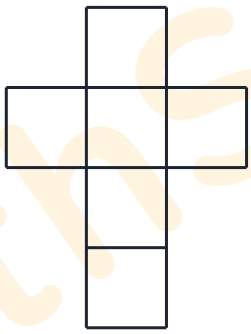
- What we see of a solid from one direction is its **profile**. The same solid can have very different profiles from different viewpoints.

Solid	One profile	Another profile
Cylinder	Rectangle (from the side)	Circle (from the top)
Cone	Triangle (from the side)	Circle (from the top)
Triangular prism	Rectangle	Triangle
Cone with its top cut off	Trapezium	Circle
Pentagonal prism	Pentagon	Rectangle

- A profile does not fix the solid: a sphere, a cylinder and a cone can all show a circle.

7 Nets

- A **net** is a flat shape that can be folded to make a solid. It is what you get when the solid is unfolded onto a plane.
- A **cube** has 11 different nets (nets that differ only by a turn or a flip count as the same). Each has 6 squares, but not every arrangement of 6 squares folds into a cube.



One of the 11 nets of a cube

- **Cuboid** $l \times b \times h$: six rectangles in three matching pairs.
- **Regular tetrahedron**: 4 equilateral triangles; it has 2 different nets. **Square pyramid**: a square with a triangle on each side.
- **Cylinder**: two circles and a rectangle. The rectangle has one side equal to the height and the other equal to the circumference $2\pi r$ of the circle.
- **Cone**: a circle and a sector of a larger circle. The radius of the sector is the slant height of the cone.
- **Octahedron**: 8 equilateral triangles; like the cube, it has 11 nets. A dodecahedron has 43,380 nets.
- A sphere has no net: a flat sheet cannot wrap a ball without wrinkles or gaps.

Example 7.1 A cylinder has radius 7 cm and height 10 cm. What are the sides of the rectangle in its net? (Take $\pi = \frac{22}{7}$.)

SOLUTION

One side is the height. The other is the circumference $2 \times \frac{22}{7} \times 7$.

10 cm and 44 cm

Example 7.2 Six squares are joined in a straight row. Is this a net of a cube?

SOLUTION

When folded, the row wraps round only 4 side faces and then overlaps. The top and bottom stay open.

No

8 Shortest paths on a cuboid

- On a plane the shortest path between two points is the straight segment joining them.
- A path on the surface of a cuboid becomes a path of the **same length** on its net. So unfold the cuboid, join the two points by a straight line and find its length, often with the Baudhāyana–Pythagoras theorem.
- The straight line must stay inside the net. Different ways of unfolding give different lengths, so **compare all the possible unfoldings** and take the smallest.

Example 8.1 An ant sits at the centre of the front face of a cube of side 4 cm. A grain of sugar is at the centre of the top face. Find the shortest path along the surface.

SOLUTION

Unfold the top face upward. The ant is 2 cm below the shared edge and the sugar is 2 cm above it, on the same vertical line.

4 cm

Example 8.2 A cuboid is 6 cm long, 3 cm wide and 2 cm high. An ant at a bottom corner wants to reach the opposite top corner. Find the shortest path along the surface.

SOLUTION

The three unfoldings give $\sqrt{(6+3)^2+2^2}=\sqrt{85}$, $\sqrt{(6+2)^2+3^2}=\sqrt{73}$ and $\sqrt{(3+2)^2+6^2}=\sqrt{61}$.

$\sqrt{61}$ cm, about 7.8 cm

9 Projections

- The **projection** of a point P on a plane is the point O of the plane such that OP is perpendicular to the plane. The projection of an object is made of the projections of all its points.
- The projection of a line segment is **never longer** than the segment. It has the same length only when the segment is parallel to the plane, and it shrinks to a point when the segment is perpendicular to the plane.
- **Parallel lines project to parallel lines.** So the projection of a parallelogram is always a parallelogram (or a line segment).
- Different objects can have the same projection, so one projection is not enough. We use three perpendicular planes:

View	Plane	Seen from
Front view	Vertical plane	The front
Top view	Horizontal plane	Above
Side view	Side plane	The side

- When sunlight falls perpendicular to a wall or floor, the **shadow** of an object is the same as its projection. A torch held close gives a larger, stretched shadow.

Solid (standing upright)	Front view	Top view	Side view
Cube	Square	Square	Square
Cuboid	Rectangle	Rectangle	Rectangle
Cylinder	Rectangle	Circle	Rectangle
Cone	Triangle	Circle	Triangle
Square pyramid	Triangle	Square with its diagonals	Triangle

Example 9.1 Three identical cubes are placed in an L shape: two side by side on the ground and one on top of the left cube. Describe the front, top and side views.

SOLUTION

Front: two squares in a row with a third square above the left one (an L shape).

Top: two squares side by side. Side: two squares, one above the other.

L shape; a 2 by 1 rectangle; a 1 by 2 rectangle

Example 9.2 A stick 10 cm long makes a projection 6 cm long on the floor. How much higher is one end than the other?

SOLUTION

The stick, its projection and the height difference form a right triangle: $h^2 = 10^2 - 6^2 = 64$.

8 cm

10 Isometric drawings

- In an **isometric projection** of a cube, all 12 edges appear with **equal length**. This happens when the cube is viewed along a line through opposite corners. Its outline is a regular hexagon.
- An **isometric grid** is made of dots (or lines) forming equilateral triangles. It has three directions: vertical for **height**, and two slanting directions for **length** and **depth**.
- To draw a solid: move along the three directions, counting units edge by edge. Parallel edges of the solid stay parallel in the drawing, and equal lengths stay equal.
- Draw faintly first, then darken only the edges that are visible.
- Four cubes joined face to face can be arranged flat in 5 ways (the "tetris" shapes) and in 3 more ways that are not flat, giving 8 solids in all.
- Some drawings on an isometric grid, such as the "impossible triangle", cannot be built as real solids. Each corner looks correct, but the whole does not fit together.

11 Common mistakes

- Thinking that the area of the Sierpinski Carpet at Step n is $1 - \frac{n}{9}$. Each step multiplies the area by $\frac{8}{9}$.
- Counting the base of a pyramid twice, or forgetting the two end faces of a prism.
- Assuming any six joined squares make a cube net.
- Taking the first unfolding as the shortest path without checking the others.
- Confusing the top view with the front view.
- Drawing an isometric cube with horizontal edges. Only the height direction is along a grid line that is vertical; the other two slant.

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